

Student Takeaway for Sync Session

<< Week 15: Probabilistic models for classification - I >>

This guide is your roadmap to making the most of our online session. Packed with essential tips and strategies, it's designed to keep you engaged, prepared, and ready to dive into a smooth and productive learning journey. Get ready to participate, learn, and thrive!

Session Overview

Session title	Probabilistic models for classification - I	
Session duration	1 hour 30mins	
Session type	Lectures: Understanding key metrics and visualisation tools for evaluating ML classification models. Case Studies: Hands-on demos on confusion matrix, ROC-AUC, PR curves, decision boundaries, and evaluation with KNN.	
Scope	This session introduces essential tools and metrics for evaluating classification models. It covers: - Key evaluation metrics and when to use them - Importance of understanding model generalisation - Tools to visualise model performance and decision boundaries - Real-world evaluation using classification reports and visualisation tools	
Learning objectives	Objective	Core capability
	Understand why visual data is critical in AI/ML	Awareness of modern data trends
	Explore industry use cases for image/video classification	Application of deep learning tools
	Use real-time tools and frameworks for video analytics	Practical understanding of real-time systems
	Apply captioning models to enhance accessibility	Hands-on with vision-language models
	Evaluate model performance using standard metrics	Analytical skill in assessing AI models
Software/Tools	- Python libraries: sklearn, matplotlib, mlxtend, numpy - IDEs: Jupyter Notebook - Demo Notebooks: Decision_Boundaries.ipynb, KNN_Classifier.ipynb	

Pro tips for success:

Think Beyond Accuracy: Metrics vary by context (e.g. imbalanced classes).

Visualise Often: Use ROC, PR curves, and boundary plots to interpret models.

Play with Parameters: Try different k-values in KNN to observe results.

Link Metrics to Applications: Choose based on real-world cost of errors.

Session Details

Slide(s)	Topic/Issue	What to expect	Notes
1-3	Introduction to Model Evaluation	Overview of model evaluation purpose Metrics, context, and multiple perspectives	Focus on generalisation and metric selection Think about the right metric for your problem
4	Confusion Matrix	Basis of all classification metrics	Understand TP, TN, FP, FN clearly
5	Accuracy	Basic correctness	Not ideal for imbalanced data
6	Precision	Correctness of positive predictions	Important when false positives are costly
7	Recall	Coverage of actual positives	Useful when false negatives are critical
8	F1 Score	Balance between Precision and Recall	Harmonic mean for imbalanced data
9	Specificity	How well negatives are detected	Complements recall
10	ROC Curve	TPR vs FPR	Helps visualise threshold trade-offs
11	ROC Plotting	Create ROC using matplotlib	Learn curve shape and AUC line
12	AUC Score	Area under ROC curve	Summary measure of model quality
13-14	PR Curve	Precision vs Recall	Great for imbalanced classification
15	Log Loss	Penalises overconfident predictions	Lower log loss = better model
16	MCC	Balanced accuracy for binary class	+1 perfect, 0 random, -1 total disagreement
17	Cohen's Kappa	Measures agreement beyond chance	Adjusts for random prediction agreement
18	Balanced Accuracy	Mean recall across classes	Handles class imbalance

Slide(s)	Topic/Issue	What to expect	Notes
19	Summary	Quick recap of metrics	Know when and where to use them
20-21	Decision Boundaries Intro	How models split feature space	Visual explanation of classifier behavior
22	Decision Boundary Definition	Prediction shift points	Line (linear) or curve (non-linear)
23	Linear Boundaries	Straight-line classifiers	Logistic regression, SVM
24	Non-Linear Boundaries	Complex contours	KNN, kernel SVM
25	Boundary Visualisation	Colour-coded classifier regions	Use <code>plot_decision_regions()</code>
26	Complexity Effects	Simplicity vs. overfitting	Tight curves may generalise poorly
27	Overfitting	Model captures training noise	Leads to erratic test behaviour
28	Classifier Comparison	Decision boundaries per model	Logistic vs. Tree vs. KNN
29-30	Feature Transformation	Impact on boundary shape	Use of polynomial features
31-32	Demo: Boundary Plots	Visualising different classifiers	Use demo notebook: <code>Decision_Boundaries.ipynb</code>
33-34	Demo: KNN Classifier	Real dataset evaluation	Try different k, check accuracy
35	Q&A	Open discussion	Clarify confusion on metrics or plots

Post session

Reflection challenge	Which metric surprised you most in terms of how it changed model evaluation? Why?
Explore more	• Read: Articles on why accuracy can be misleading • Watch: Tutorials on ROC/PR curves using sklearn • Experiment: Run demo notebooks with your own data
Get inspired	Did you know an F1 score can save lives? It's used in medical diagnosis models to ensure false negatives are minimal. Your smart choice of metrics could impact real-world outcomes.
The journey ahead	Next, we explore advanced ML applications like Natural Language Processing, Recommendation Engines, and AI-powered decision-making systems.